



INSTITUTO SUPERIOR TÉCNICO

DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

FORENSICS CYBER-SECURITY

MEIC, METI

Lab Assignment I

OPERATION DEEPFISH – Stage I

2026/2027

nuno.m.santos@tecnico.ulisboa.pt

Introduction

OPERATION DEEPFISH is a three-stage digital-forensics investigation conducted by your team. Each stage is introduced through a separate laboratory assignment and extends the evidence available in the case. This document introduces Stage I, in which you will act as an independent digital-forensics investigator and assess the incident based on the material made available for the case. Your objective is to identify and examine relevant digital evidence, reconstruct the activity where possible, and support your conclusions through documented and reproducible analysis. The assignment provides practical experience in suspicious-message analysis, digital-file analysis, and, where supported by the evidence, steganalysis or related concealment techniques.

To perform this assignment, prepare a controlled environment for artifact analysis. We recommend using a Kali Linux virtual machine, although equivalent tooling may be used. Preserve original artifacts whenever possible, work on copies, and document the steps required to reproduce your results.

Case Background

Between 19 and 22 August 2026, several members of the IST teaching staff received e-mail messages similar to the one shown in Figure 1. The messages appeared to originate from the *Serviços de Informática* (SI) and informed recipients that their Técnico ID passwords were about to expire. Recipients were instructed to follow an embedded link to change their passwords.



Figure 1: Example of a reported password-expiration message. Recipient-identifying information has been redacted for privacy and data-protection purposes.

The volume and recurrence of the messages raised concerns among the recipients. Several professors independently reported the messages to the SI and questioned whether the password-expiration notices

and the embedded link were legitimate. Some recipients had already followed the link before reporting the messages. These independent reports prompted an initial internal response.

The message reproduced in Figure 1 is a recipient-side capture submitted with one of these reports. Recipient-identifying information was subsequently redacted solely for privacy and data-protection purposes. During its initial triage, the SI compared the reported messages and recorded the destination of the embedded “here” hyperlink. Every reported copy examined before the case was referred for forensic analysis pointed to the same destination: <https://fenix.dpss.inesc-id.pt/index.html>. This consistency supported treating the reports as instances of the same incident, potentially reflecting an ongoing phishing campaign.

Forensic Assignment

Initial triage was performed by the *Núcleo de Redes e Cibersegurança* (NRC), with support from the *Núcleo de Sistemas* (NS). Your group has been hired to act as the forensic team supporting NS.

The Stage I referral package consists of the redacted recipient-side capture shown in Figure 1 and the common destination URL recorded during the initial triage. It does not include a raw e-mail export or any mail-system records. Your initial forensic examination is therefore grounded in what was visible to the recipient and in the web resource referenced by the message. At this stage, neither the origin nor the purpose of the reported activity has been established.

Your mission is to determine whether the reported e-mails are part of a scam, characterize the nature of the attack if one exists, and attempt to attribute responsibility. The investigation should begin with the redacted e-mail itself, particularly the references it contains, and should carefully follow any leads that emerge from the available evidence.

During the investigation, you may encounter additional digital materials, such as websites or files hosted online. These artifacts must be treated as potential evidence and examined thoroughly to support your conclusions. If you find indications of steganography, obfuscation, or other concealment techniques, your team should be prepared to investigate them and recover any hidden material.

Your investigation will be evaluated based on two complementary components: the forensic examination and analysis performed, and the drawn conclusions in response to the case questions.

Investigation and evidence analysis. Conduct an impartial and technically supported examination of the material provided and relevant evidence identified during the investigation. Material findings must be supported by appropriate technical analysis and documented in sufficient detail for the basis of each finding to be understood and, where applicable, reproduced. The provenance and handling of relevant evidence must be documented in accordance with the forensic working requirements set out below.

Questions to answer. Based on the evidence examined during the investigation, your report must explicitly answer the following questions. Treat the suspicions expressed in the reports as investigative context rather than established fact, and clearly distinguish conclusions supported by evidence from matters that remain inferential.

1. **Nature of the incident.** (a) Did you find evidence that the reported activity was part of a scam? If so, describe the nature of the scam and reconstruct how it was conducted. (b) Did you uncover any additional concealed artifacts during your investigation? If so, explain how these artifacts were concealed and describe the methodology used to recover them. For both (a) and (b), support your conclusions with the relevant evidence identified during the investigation.
2. **Interpretation of the evidence.** Focusing on the recovered concealed material, can you identify relationships among the recovered items, explain what they may represent, and formulate a supported hypothesis about their significance to the case? Clearly identify which parts of your interpretation are established by evidence and which remain inferential.

3. **Attribution and next investigative steps.** Does the available evidence support any conclusion about the identity of the person or persons responsible for the scam or any other suspicious activity uncovered? Discuss the strengths and weaknesses of the attribution evidence you have found. Provide recommendations to SI regarding the next investigative steps, including any additional evidence that should be preserved or obtained before further action is taken.

Requirements. The quality of the investigation depends not only on its final conclusions but also on its evidentiary basis, reproducibility, integrity, and temporal reconstruction. At a minimum:

- preserve original evidence whenever possible; perform destructive or modifying operations on copies; and calculate and report SHA-256 hashes for every recovered concealed payload for which a stable, byte-exact representation exists, as well as for other relevant recovered or preserved files where appropriate;
- record the provenance of relevant artifacts, the path by which they were discovered or recovered, and the investigative steps followed to identify, access, and examine relevant evidence sources;
- document the commands, tools, scripts, and material analysis steps required to reproduce key findings. Where concealment, obfuscation, or steganography is suspected, document what prompted that suspicion and how it was tested;
- reconstruct the relevant chronology from the available evidence, using timestamps carefully, including time-zone information where available, and state any normalization, conversion, assumptions, or temporal uncertainty involved;
- clearly distinguish observed facts, forensic inferences, and hypotheses, ensuring that every material finding or conclusion can be linked to one or more evidence items.

Deliverables

Submit a single ZIP archive through Fenix by **23 September 2026**. The archive must contain the following four deliverables:

- **Digital forensic report.** A report documenting the investigation and evidence analysis performed in accordance with the requirements specified in the **Forensic Assignment** section. The report must explicitly answer all case questions and demonstrate compliance with the **Requirements**. Each material finding or conclusion must be supported by identified evidence and by a sufficiently documented and reproducible analysis process. Use the report template available on the course website unless instructed otherwise.
- **Evidence artifacts.** The relevant evidence artifacts recovered or preserved during the investigation. The report must identify each submitted artifact by file name and SHA-256 hash and state its source or provenance.
- **Auxiliary items.** Any scripts, programs, command outputs, notes, or other supporting material required to reproduce or substantiate important findings.
- **Timeline.** A timeline of the most relevant events in the case. Each event must be supported by evidence and identify the source of its timestamp, together with any assumptions or time-zone conversions applied. Use the timeline template available on the course website unless instructed otherwise.

Lab hint (not case evidence): The provided evidence set contains a total of **8 hidden secrets**.

Use of Artificial Intelligence

The use of Artificial Intelligence (AI) tools is permitted in this assignment, provided that it complies with the applicable Instituto Superior Técnico guidelines for the responsible use of AI.

Students remain fully responsible for all submitted work, including findings, technical conclusions, scripts, commands, procedures, and any other material produced with the assistance of AI tools. AI-generated or AI-assisted content must be independently verified for correctness, validity, and relevance.

Students must understand and be able to explain, justify, and, where requested, reproduce or demonstrate the investigation they submit, including the procedures followed, the tools and techniques used, the evidence obtained, and the conclusions presented. Failure to do so may be taken into account when assessing whether the student has demonstrated the competencies evaluated by this assignment.

Any use of AI tools must be declared in accordance with the applicable Técnico guidelines, identifying the tools used and the purposes for which they were used. Students should consult the guidelines and regulations published by the Conselho Pedagógico at: <https://conselhopedagogico.tecnico.ulisboa.pt/regulamentos/faqs-regulamento-de-avaliacao-2026/>

The Fundação para a Ciência e a Tecnologia (FCT), through FCCN, also provides the **IAedu** platform to the Portuguese higher-education and research community. The service is available to Técnico students using their institutional credentials and provides access to several AI models without requiring an individual paid subscription. Students wishing to use AI tools in this assignment are encouraged to use this institutional service: <https://si.tecnico.ulisboa.pt/servicos/servicos-externos/iaedu/>

Good luck!